LRCS-Mamba: Low-Rank Mamba With Content-Aware Sparse Scanning for Hyperspectral Image Classification

Jinliang An, *Member, IEEE*, Longlong Dai, *Graduate Student Member, IEEE*, Dengru Han, Weidong Zhang, *Senior Member, IEEE*, and Xiangrong Zhang, *Senior Member, IEEE*

***Abstract*—Recently, state space models (SSMs), particularly the Mamba architecture, have emerged as a powerful paradigm for hyperspectral image (HSI) classification due to their linear complexity and ability to capture long-range dependencies. However, existing Mamba-based approaches mainly rely on predefined geometric scanning patterns that ignore spatial heterogeneity, leading to spatial–semantic misalignment. Moreover, standard full-rank parameterization in SSMs fails to exploit the intrinsic low-rank structure of HSI. Without structural constraints, redundant state dynamics may propagate noisy responses caused by misaligned input sequences, further limiting the robustness of spectral–spatial representation learning. To address these coupled issues, we propose LRCS-Mamba, a dual-branch Mamba network with content-aware sparse scanning (CASS) and low-rank state-space modeling for HSI classification. Specifically, we design a CASS mechanism that constructs input sequences via center-neighbor spectral affinity and Top-k selection, reducing heterogeneous boundary-pixel contamination before state evolution. To reduce noisy responses during state evolution, we further develop a low-rank SSM (LRSSM) that factorizes the core evolution matrices and constrains state dynamics within a compact latent subspace. Finally, a lightweight convolutional neural network (CNN) branch and an adaptive dual-branch sparse gating (ADSG) module are integrated to balance global semantics and local textures. Extensive experiments on four benchmark datasets demonstrate that LRCS-Mamba consistently outperforms state-of-the-art methods and achieves robust HSI classification performance.**

***Index Terms*—Hyperspectral image (HSI) classification, low-rank manifold, Mamba, sparse scanning, state-space model (SSM).**

I. Introduction

HYPERSPECTRAL image (HSI) contains hundreds of continuous narrow spectral bands, which form a 3-D data cube containing rich spatial geometry and fine-grained spectral characteristics. This unique integration of spatial and spectral information enables HSI to demonstrate significant potential in various fields [1]. However, the inherent high-dimensional spectral redundancy and complex spatial heterogeneity of HSI pose challenges for HSI classification. A core difficulty in this field remains how to extract compact and discriminative features from massive redundant data while effectively suppressing band noise.

Over the past decade, the evolution of deep learning has significantly advanced HSI classification. Early research primarily relied on convolutional neural networks (CNNs), evolving from 1D-CNNs for spectral feature extraction to 3D-CNNs for joint spectral–spatial modeling [2]. However, constrained by the fixed receptive fields of convolutional kernels, CNNs struggle to capture long-range spatial dependencies. Subsequently, vision transformers (ViTs) addressed global modeling through self-attention mechanisms, yet their $O(N^2)$ computational complexity becomes prohibitive when processing high-resolution HSI. To handle the irregular distribution of land-cover objects in HSI, graph neural networks (GNNs) [3] were introduced to aggregate neighborhood information by constructing topological graphs. Although GNNs excel at processing non-Euclidean data, their intensive adjacency matrix computations and complex graph construction processes limit inference efficiency.

Recently, structured state-space models (SSMs), particularly the Mamba architecture [4], have emerged as a promising foundation model paradigm. With its linear computational complexity and global receptive field advantages, Mamba has rapidly become a research hotspot. The core innovation of Mamba lies in the selective scan mechanism, which utilizes data-dependent parameterization to allow the model to dynamically filter key information from continuous signals. Preliminary studies such as MambaHSI [5] and 3DSS-Mamba [6] have demonstrated that Mamba is exceptionally well-suited

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Jinliang An, Longlong Dai, and Dengru Han are with the School of Information Engineering, Henan Institute of Science and Technology, Xinxiang 453003, China (e-mail: anjinliang@stu.xidian.edu.cn; dailonglong@stu.hist.edu.cn; handengru@stu.hist.edu.cn).

Weidong Zhang is with the School of Electrical and Information Engineering, Zhengzhou University, Zhengzhou 450001, China, and also with the School of Information Engineering, Henan Institute of Science and Technology, Xinxiang 453003, China (e-mail: zwd_wd@163.com).

Xiangrong Zhang is with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: xrzhang@mail.xidian.edu.cn).

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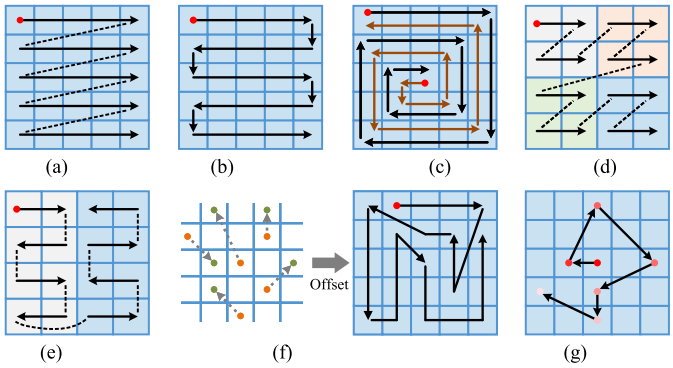


Fig. 1. Comparison of different spatial scanning strategies. (a) Vim [11] uses a Z-shaped scan to flatten 2-D features into 1-D sequences, disrupting spatial continuity and locality. (b) Zigma [12] adopts an S-shaped path to improve continuity but overlooks local neighborhood consistency. (c) ABS-Mamba [13] applies alternating bidirectional scans to enhance global context, at the cost of mixing distant semantics. (d) LocalMamba [14] employs window-based scanning to preserve locality, while Z-shaped traversal breaks continuity across windows. (e) MaIR [15] performs stripe-wise S-shaped scanning, balancing locality and spatial continuity. (f) DefMamba [7] adopts deformable scanning to adaptively focus on informative regions, but remains geometry-driven without explicit spectral modeling. (g) Ours adopts a CASS strategy that jointly preserves local coherence and long-range spatial dependencies.

for processing HSI with continuous spectral characteristics, achieving superior performance to Transformers while maintaining a low memory footprint.

Despite the rapid development of Mamba-based HSI classification methods, existing research mostly relies on directly adopting standard modules from natural language processing (NLP) or computer vision (CV). These approaches often neglect the unique physical properties and spatial distribution patterns of HSI, leading to significant limitations in complex scenarios. Specifically, two core issues remain unresolved.

1) *Spatial–Semantic Misalignment in Sequence Construction:* Existing Mamba-based HSI methods usually flatten HSI features into 1-D sequences through predefined, multidirectional, grouped, or geometry-driven scanning paths [7]. Fig. 1 provides an intuitive comparison of representative scanning strategies, illustrating that most existing paths primarily reshape spatial adjacency rather than explicitly selecting semantically related neighbors. Since land-cover objects in HSI often have irregular shapes and complex boundaries, such geometry-driven serialization may connect spatially adjacent but semantically different pixels, or separate pixels from the same homogeneous region. Motivated by sparse representation theory, effective feature interaction should focus on the most informative neighbors rather than passively accepting all fixed spatial neighbors. Therefore, sequence construction should be guided by center-neighbor content affinity and sparse selection, rather than only by fixed spatial adjacency [8].
2) *Redundant Full-Rank State Evolution Inconsistent With the Low-Dimensional Structure of HSI:* HSI usually contains strong interband correlations and is commonly assumed to lie on compact low-dimensional spectral–spatial manifolds [9]. Nevertheless, most existing Mamba-based HSI methods still employ standard full-rank SSM operators or simple channel grouping strategies [10]. Such parameterization may introduce redundant state dynamics and make the model sensitive to noisy spectral variations, especially under limited labeled samples. Meanwhile, existing low-rank or sparse representation methods mainly impose structural constraints on data matrices, feature projections, or attention maps, rather than embedding the low-rank prior into the state evolution process of Mamba. Therefore, it is desirable to introduce low-rank constraints directly into the core SSM operators, so that state dynamics can evolve within a compact latent subspace consistent with the intrinsic structure of HSI.

Critically, these two issues are not independent. Geometry-driven scanning may introduce heterogeneous boundary pixels into the sequence, and a full-rank SSM, lacking structural constraints, propagates these noisy signals through unconstrained state dynamics instead of explicitly constraining them. Input-level misalignment is thus reinforced by modeling-level redundancy, and addressing both sides jointly is important.

Motivated by the discussion above, we develop a dual-branch Mamba framework with content-aware sparse scanning (CASS) and low-rank state-space modeling, termed LRCS-Mamba, for HSI classification. The framework contains a local texture perception branch (LTPB) and a global semantic perception branch (GSPB). The local branch uses lightweight depthwise pointwise separable convolutions to capture fine-grained spatial textures, edge structures, and boundary-sensitive details. The global branch first constructs semantically consistent sequences through CASS, where neighboring pixels are selected according to center-neighbor spectral affinity and Top-k sparsification. The selected sequence is then processed by a low-rank Mamba block (LRMB), in which the core SSM operators are factorized to constrain state evolution within a compact latent manifold. Finally, an adaptive dual-branch sparse gating (ADSG) module integrates local texture cues and global semantic representations for final classification.

Compared with existing Mamba-based HSI classifiers that mainly modify scanning routes or directly adopt full-rank SSM operators, LRCS-Mamba targets the coupled problem between sequence construction and state evolution. Geometry-driven scanning may introduce heterogeneous boundary pixels into the sequence, while unconstrained full-rank state evolution may further propagate such noisy responses. Therefore, the core novelty of LRCS-Mamba lies in jointly improving what to scan and how to model the scanned sequence. CASS constructs content-aware sparse sequences according to center-neighbor spectral affinity, while low-rank SSM (LRSSM) embeds the low-rank HSI prior into the core SSM operators. Together with the local texture branch and adaptive gated fusion, LRCS-Mamba forms a unified framework for local-global complementary spectral–spatial representation learning.

The main contributions of this work are summarized as follows.

1) We propose a scanning strategy that constructs Mamba input sequences via center-neighbor spectral affinity

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and Top-k selection. Guided by sparse representation theory, CASS selects semantically related neighbors and reduces heterogeneous boundary-pixel contamination before state evolution, thereby addressing the input-level spatial–semantic misalignment.

2) We design a low-rank state-space operator that factorizes the core SSM matrices, constraining state dynamics within a compact latent subspace. Unlike conventional low-rank methods that constrain feature representations or attention maps, LRSSM embeds the low-rank prior into Mamba state evolution. Coupled with CASS, it helps suppress residual boundary interference from being amplified by unconstrained full-rank dynamics.
3) We build a dual-branch framework with learnable sparse gating that fuses local CNN textures and global Mamba semantics while suppressing low-contribution responses through threshold-based sparsification. This design integrates what to scan, how to model, and how to fuse, enabling robust HSI classification under limited labels and complex boundaries.

The remainder of this article is organized as follows. Section II systematically reviews related work. Section III presents the proposed LRCS-Mamba in detail. Section IV demonstrates experimental results, and Section V concludes the article and discusses directions for future research.

II. Related Work

A. Deep Learning Architectures for HSI

With their strong local feature extraction ability, CNNs have been widely adopted for HSI classification. Hu et al. [16] pioneered the use of 1-D CNNs for spectral feature learning. Subsequently, 2-D CNNs and 3-D CNNs [17], [18] were widely explored to jointly model spectral–spatial information. Later, various optimization strategies, such as multiscale designs and multibranch architectures, were broadly incorporated into CNN-based frameworks to further improve classification accuracy [19]. However, despite their strong performance, the inherently local receptive fields of convolutional kernels limit CNNs in capturing long-range spatial dependencies across the scene. Simultaneously, considering that the spectral continuity of HSI data is analogous to time series, recurrent neural networks (RNNs) [20] have been employed to process spectral sequences. RNNs effectively capture dependencies between bands through gating mechanisms. However, the sequential nature of RNNs makes training difficult to parallelize, resulting in low inference efficiency. Moreover, they still face challenges like gradient vanishing or forgetting early information when processing extremely long sequences.

To overcome the locality of CNNs, Transformer-based methods were introduced to HSI classification [21]. By leveraging the self-attention mechanism, Transformers can establish global pixel-to-pixel dependencies. To improve spectral–spatial representation efficiency, QTN [22] incorporates quaternion algebra into the Transformer framework to explicitly model the intrinsic coupling among spectral bands, achieving more compact and structured feature representations. ISSFormer [23] further enhances Transformer-based HSI classification by parallelizing global spectral attention and local spatial perception, demonstrating that coordinated spectral–spatial interaction is crucial for accurate hyperspectral scene understanding. However, the computational complexity of self-attention grows quadratically with the number of pixels, imposing a heavy computational and memory burden that limits its application in large-scale HSI scenarios.

Moreover, land-cover objects in HSI are often complex and irregularly distributed, posing challenges for single-network architectures. To fully exploit the complementary strengths of these networks, researchers have proposed various hybrid architectures [24]. The fusion of CNN and graph convolution network (GCN) frameworks first extracts local spectral–spatial features with CNNs and then models structural information using GCNs. S3F2Net [25] effectively combines local details with topological structures. The fusion of Transformer and GCN combines global dependency modeling of Transformers with the structure-aware capability of GCNs, enhancing performance in complex scenes [26]. The combination of Transformer and CNN extracts local textures with CNN and captures long-range dependencies with Transformers, achieving a local–global complementarity [27]. These hybrid approaches significantly improve feature representation in hyperspectral classification, yet efficiently leveraging structural information while maintaining computational efficiency remains an important challenge.

B. SSMs and Mamba

Structured SSMs [28] provide a principled framework for modeling long sequences by discretizing continuous dynamical systems. Benefiting from selective state updates and hardware-aware parallelization, the Mamba architecture achieves linear computational complexity while preserving a global receptive field comparable to Transformers [29]. Recent studies have begun adapting Mamba to HSI classification by embedding spectral–spatial priors into the state-space modeling framework. SS-Mamba [30] represents early attempts to tailor Mamba to hyperspectral data through decoupled spectral and spatial modeling, demonstrating its potential as an efficient alternative backbone for HSI classification. Beyond these basic adaptations, several works further enhance Mamba-based architectures by incorporating domain-specific inductive biases. MorpMamba [31] integrates morphological operations into the Mamba framework to better preserve structural and edge information inherent in HSI data. In addition, semi-Mamba [32] extends Mamba-based state-space modeling to semisupervised multimodal remote sensing scenarios, highlighting its broader applicability beyond fully supervised HSI classification.

In recent years, researchers have developed various enhanced scanning mechanisms to better preserve spectral–spatial semantics and reduce redundancy. Local Enhanced Mamba [33] integrates multidirectional scans to combine local and global contexts, enhancing spatial discrimination. Yao et al. [34] proposed SpectralMamba, which compresses continuous spectra into shorter sequences while maintaining both short- and long-term contextual

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information. He et al. [6] introduced 3DSS-Mamba, performing pixel-wise scanning along both spectral and spatial dimensions and constructing multiple scanning paths to capture global spectral–spatial relationships. He et al. [35] further proposed IGroupSS-Mamba, which partitions high-dimensional features into nonoverlapping groups with specific scanning directions, enabling hierarchical and multidirectional context modeling. Beyond geometric or grouped scanning, S^{2}Mamba [36] adopts patch-level cross scanning with bidirectional spectral modeling to jointly capture spatial–spectral dependencies, while Mamba-in-Mamba [37] employs a centralized cross-scan during image-to-sequence transformation to facilitate structured tokenization and multiscale semantic aggregation for HSI classification. Moreover, hybrid approaches like S2GL-MambaResNet [38] combine global–local Mamba scanning with residual networks to capture multiscale spectral–spatial context. Collectively, these studies indicate a clear trend toward content-adaptive, multidirectional, and semantically aware scanning strategies, which effectively balance local continuity and global context. However, land-cover boundaries in HSIs remain highly irregular, and even advanced scanning paths may still introduce context aliasing when semantically distinct pixels are forced into the same sequence.

To synthesize the advantages of different architectures, researchers have proposed various hybrid models. To balance local textures with global semantics [39], PHDMamba [40] and DualMamba [4] utilize parallel or serial combinations of CNNs and Mamba. Furthermore, to handle irregular structures, GraphMamba [41] and DBMGNet [42] attempt to fuse GCNs with Mamba. Although these hybrid models improve performance, the inclusion of GCNs often reintroduces heavy explicit graph computations, which partially offsets Mamba's original design intent of being lightweight and efficient. Therefore, exploring an implicit yet efficient method for graph structure modeling is particularly important.

C. Low-Rank and Sparse Representation Learning in Deep Learning

The core challenge of HSI processing lies in the massive data volume and high spectral correlation [43]. Utilizing low-rank and sparse prior knowledge to recover intrinsic structures from high-dimensional data has long been a theoretical cornerstone of HSI analysis [44]. Zhuang et al. [45] and Zeng et al. [46] pioneered methods based on low-rank sparse matrix decomposition (LRSD) for denoising and anomaly detection, proving that modeling spectra on a low-dimensional manifold effectively suppresses noise and enhances discriminative power. These traditional optimization methods provide a strong physical inductive bias for deep network design. Inspired by these physical priors, researchers have begun to integrate low-rank and sparse constraints into deep neural networks to reduce model complexity while enhancing feature robustness. GLB-LRA [47] jointly models global and local low-rank structures to enhance intraclass consistency and interclass separability, providing strong empirical evidence for the effectiveness of low-rank priors in HSI classification. In the Transformer domain, to alleviate the computational complexity of self-attention, tensor product attention [48] utilizes tensor decomposition to parameterize the attention matrix into a low-rank form. For HSI classification, sparse spectral Transformer [49] and multiscale low-rank transformer [50] introduce sparse attention masks or low-rank projections, successfully maintaining global modeling capabilities while reducing parameter counts. Furthermore, recent theoretical work has revealed that the discriminative power of Softmax attention stems from the sparsity and low-rankness of the matrix, providing theoretical support for designing more efficient sequence models [51]. Despite the success of low-rank and sparse theory in Transformers, existing Mamba methods typically perform full-rank modeling on high-dimensional spectra or use simple hard grouping. This fails to fully exploit the intrinsic low-rank manifold of HSI.

III. Proposed Method

A. Overall Architecture

As illustrated in Fig. 2, LRCS-Mamba consists of three main stages: shallow feature embedding, dual-branch deep feature extraction, and adaptive gated fusion. Given an original HSI $\mathbf{X} \in \mathbb{R}^{h\times w\times d}$, where h and w denote the spatial dimensions and d is the number of spectral bands, we first extract 3-D patches of size $p \times p \times d$ in a pixel-wise manner. Each patch is processed by a shallow feature embedding module, which projects the original spectral dimension into a compact c-channel feature space, resulting in a representation of size $p \times p \times c$. The embedded features are then fed into a parallel dual-branch structure composed of a LTPB and a GSPB. The LTPB employs lightweight depthwise pointwise separable convolutions with normalization and nonlinear activation to capture fine-grained local textures, edge structures, and boundary-sensitive spatial patterns, producing the local feature $\mathbf{F}_{\text{local}}$. In contrast, the GSPB serves as the core semantic modeling branch. It consists of the CASS strategy and the LRMB. CASS dynamically selects semantically homogeneous neighbors according to center-neighbor content affinity, thereby constructing compact and semantically consistent input sequences for Mamba. The resulting sequence is further processed by LRMB, where LRSSM introduces low-rank structural constraints into the core SSM operators to encourage compact spectral–spatial dependency modeling in a latent subspace, producing the global feature $\mathbf{F}_{\text{global}}$. Finally, $\mathbf{F}_{\text{local}}$ and $\mathbf{F}_{\text{global}}$ are fused by the ADSG module, which adaptively balances the contributions of local details and global semantics through learnable thresholds. The fused representation $\mathbf{F}_{\text{fused}}$ is then fed into global average pooling (GAP) and a Softmax classifier for final land-cover prediction.

B. Local Texture Perception Branch

The LTPB is designed to capture fine-grained spatial textures and local structural variations within each hyperspectral patch. It consists of two consecutive depthwise pointwise separable convolution layers with normalization and activation functions. Given the embedded local HSI patch feature $\mathbf{X}_{\text{p}} \in \mathbb{R}^{p\times p\times c}$, the first layer performs spatial

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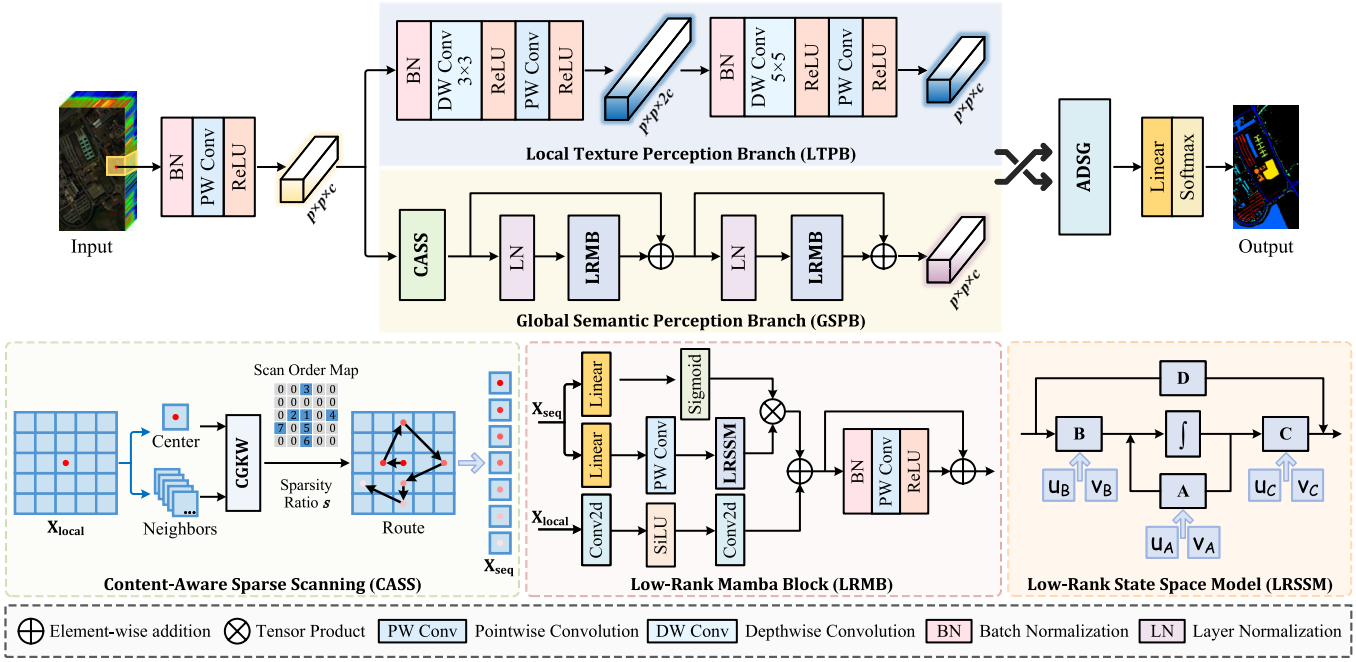


Fig. 2. Overall structure of the LRCS-Mamba framework. LRCS-Mamba adopts a dual-branch architecture to jointly exploit local textures and global semantics for HSI classification. An input HSI patch is first embedded into compact features, which are then processed by an LTPB to preserve fine-grained spatial details and a GSPB that constructs semantically consistent token sequences via CASS and captures long-range dependencies using stacked LRMBs, in which LRSSM serves as the core state-space operator. The two branch features are finally integrated by the ADSG module through adaptive sparse gating and fed into a linear classifier to produce the final prediction.

regularization using batch normalization, extracts local texture patterns with a 3×3 depthwise convolution, and expands the channel dimension from c to $2c$ through a 1×1 pointwise convolution. ReLU activation is applied after each convolution. This process is formulated as

$$\mathbf{F}_{\mathrm{c}} = \mathrm{ReLU}\left(\mathrm{PW}_1\left(\mathrm{ReLU}\left(\mathrm{DW}_{3\times3}\left(\mathrm{BN}\left(\mathbf{X}_{\mathrm{p}}\right)\right)\right)\right)\right) \tag{1}$$

where $\mathbf{F}_{\mathrm{c}} \in \mathbb{R}^{p\times p\times 2c}$ is the intermediate feature map, $\mathrm{BN}(\cdot)$ denotes batch normalization, $\mathrm{ReLU}(\cdot)$ denotes the activation function, $\mathrm{DW}_{3\times3}(\cdot)$ denotes the 3×3 depthwise convolution, and PW_1 denotes the first 1×1 pointwise convolution.

Subsequently, $\mathbf{F}_{\mathrm{c}}$ is fed into the second layer to capture broader local contextual details. This layer adopts a 5×5 depthwise convolution to enlarge the receptive field, followed by a 1×1 pointwise convolution to compress the channel dimension from $2c$ back to c for dual-branch fusion

$$\mathbf{F}_{\mathrm{local}} = \mathrm{ReLU}\left(\mathrm{PW}_2\left(\mathrm{ReLU}\left(\mathrm{DW}_{5\times5}\left(\mathrm{BN}\left(\mathbf{F}_{\mathrm{c}}\right)\right)\right)\right)\right) \tag{2}$$

where $\mathbf{F}_{\mathrm{local}} \in \mathbb{R}^{p\times p\times c}$ represents the final output of the LTPB, $\mathrm{DW}_{5\times5}(\cdot)$ denotes the 5×5 depthwise convolution, and PW_2 denotes the second 1×1 pointwise convolution. In both layers, the convolution stride is set to 1. The padding sizes of the 3×3 and 5×5 depthwise convolutions are set to 1 and 2, respectively, so that the spatial resolution remains unchanged. Through this two-stage design with progressively enlarged receptive fields, LTPB enhances local discriminability while maintaining computational efficiency via depthwise pointwise separable convolutions.

C. Global Semantic Perception Branch

The GSPB is the core semantic modeling branch of LRCS-Mamba. It is designed to capture nonlocal contextual dependencies among semantically related pixels within local HSI patches. Specifically, GSPB stacks two modeling layers, each consisting of layer normalization (LN) and an LRMB. LN stabilizes the feature distribution, while LRMB combines CASS with the LRSSM. CASS constructs semantically consistent token sequences by selecting homogeneous neighbors according to center-neighbor content affinity, and LRSSM models these sequences under low-rank structural constraints. The resulting output is denoted as the global semantic feature $\mathbf{F}_{\mathrm{global}}$.

1) Content-Aware Sparse Scanning: Although Mamba excels at sequence processing, its prevalent scanning strategies are inherently spatially rigid, which may lead to spatial–semantic misalignment at complex boundaries, inevitably introducing heterogeneous noise into the state space. To overcome this limitation, we propose CASS, which aims to dynamically construct input sequences based on the semantic affinity between pixels rather than their geometric positions.

Given a local HSI patch $\mathbf{X}_{\mathrm{p}} \in \mathbb{R}^{p\times p\times c}$, a center-anchored Gaussian kernel weighting (CGKW) is employed to measure the affinity between the center pixel and neighboring pixels. Specifically, we regard the center pixel $\mathbf{x}_c \in \mathbb{R}^c$ as an anchor point, define the set of pixels in its neighborhood as $\Omega = \{\mathbf{x}_i \mid i = 1, \ldots, p^2 - 1\}$, and construct an implicit star-shaped graph centered at $\mathbf{x}_c$. To measure the semantic association between neighboring pixels and the center pixel, we adopt a Gaussian kernel-based similarity metric. For the ith pixel $\mathbf{x}_i \in \mathbb{R}^c$ in the

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neighborhood, its affinity score S_i with the center pixel $\mathbf{x}_c$ is defined as

$$S_i = \exp\left(-\frac{\|\mathbf{x}_c - \mathbf{x}_i\|_2^2}{2\sigma^2}\right) \tag{3}$$

where $\|\cdot\|_2$ denotes the Euclidean norm, and σ is a bandwidth parameter controlling the width of the Gaussian kernel. This formula quantifies the similarity between pixels based on spectral feature distances. Since the affinity score is used only to rank neighboring pixels for Top-k selection, σ mainly affects the numerical scale of the Gaussian scores rather than their relative ranking. In our implementation, σ is fixed to 5 for all datasets.

Based on the attention-based sparsity prior theory, the feature representation of the center pixel should be primarily determined by a few key homogeneous neighbors, while excessive background context may dilute the discriminability of the features. Therefore, instead of including all neighboring pixels into the sequence, we sort the neighbors in Ω by their affinity scores S_i in descending order and select the top $(k-1)$ most relevant pixels. Let $\{i_1, \ldots, i_{k-1}\}$ denote the selected indices satisfying

$$S_{i_1} \geq S_{i_2} \geq, \ldots, \geq S_{i_{k-1}}. \tag{4}$$

To explicitly preserve the spatial correspondence of the selected tokens, we simultaneously record their original spatial indices during the selection and reordering process. Let

$$\mathbf{P}_{\text{seq}} = \left[p_c, p_{i_1}, \ldots, p_{i_{k-1}}\right] \in \mathbb{R}^k \tag{5}$$

where p_c and p_{i_j} denote the original spatial positions of the center pixel and the jth selected neighboring pixel within the $p \times p$ patch, respectively. The token gathering and reordering operation is then formulated as

$$\mathbf{X}_{\text{seq}} = \text{Gather}\left(\mathbf{X}_{\text{p}}, \mathbf{P}_{\text{seq}}\right) \in \mathbb{R}^{k \times c}. \tag{6}$$

Here, Gather$(\cdot)$ simultaneously performs token selection and reordering according to the affinity-based spatial index sequence $\mathbf{P}_{\text{seq}}$. The center pixel is placed at the first position, followed by the selected neighboring pixels in descending order of their affinity scores. The resulting $\mathbf{X}_{\text{seq}}$ is subsequently fed into the LRMB for semantic sequence modeling.

Importantly, $\mathbf{P}_{\text{seq}}$ is retained to preserve the correspondence between each sequence token and its original spatial position, enabling spatial restoration after sequence modeling. By selecting the Top-k most semantically relevant neighbors, CASS automatically suppresses spectrally distinct pixels even at complex land-cover boundaries, transforming the fixed geometric neighborhood into a dynamic semantic domain. This allows Mamba to perform state evolution along semantically consistent paths, thereby reducing semantic aliasing and improving the modeling of irregular structures.

2) Low-Rank Mamba Block: To jointly capture long-range semantic dependencies and fine-grained spatial textures, we design the LRMB. As illustrated in Fig. 2, LRMB adopts an explicit three-path design, including a global semantic path driven by LRSSM, a gating path for adaptive information modulation, and a lightweight local convolutional path for texture compensation. The three paths cooperate within a single block so that global state evolution is guided by semantically relevant information while local high-frequency details are preserved.

Given the input semantic sequence $\mathbf{X}_{\text{seq}} \in \mathbb{R}^{k \times c}$ produced by the CASS module, in the global semantic path, $\mathbf{X}_{\text{seq}}$ is first projected to reorganize channel-wise information and then mapped by a pointwise convolution to match the LRSSM operator. The resulting representation is fed into LRSSM to perform state evolution on a compact latent manifold, producing the Mamba-based feature $\mathbf{F}_{\text{mamba}} \in \mathbb{R}^{k \times c}$

$$\mathbf{F}_{\text{mamba}} = \text{LRSSM}\left(\text{PWConv}\left(\text{Linear}\left(\mathbf{X}_{\text{seq}}\right)\right)\right). \tag{7}$$

Simultaneously, to dynamically control the information flow, the gating path generates an adaptive modulation map from the same input. Specifically, it expands the channel dimension through a linear projection and produces the gating signal $\mathbf{G} \in \mathbb{R}^{k \times c}$ via the Sigmoid activation

$$\mathbf{G} = \text{Sigmoid}\left(\text{Linear}\left(\mathbf{X}_{\text{seq}}\right)\right) \tag{8}$$

where Sigmoid$(\cdot)$ is used to obtain bounded gating weights. The Mamba-based feature is then filtered by element-wise gating, which helps suppress redundant or noisy variations before fusion

$$\mathbf{F}_{\text{gate}} = \mathbf{F}_{\text{mamba}} \odot \mathbf{G} \tag{9}$$

where $\odot$ denotes element-wise multiplication. After sequence-level semantic aggregation, the sparse feature $\mathbf{F}_{\text{gate}} \in \mathbb{R}^{k \times c}$ is restored to its original spatial layout using the retained index sequence $\mathbf{P}_{\text{seq}}$. Specifically, the local patch is first flattened as

$$\mathbf{X}_{\text{flat}} = \text{Flatten}\left(\mathbf{X}_{\text{p}}\right) \in \mathbb{R}^{p^2 \times c}. \tag{10}$$

The processed features $\mathbf{X}_{\text{flat}}$ are assigned to their corresponding spatial positions and then reshaped to recover the original spatial layout

$$\begin{aligned} \mathbf{X}_{\text{flat}}\left[\mathbf{P}_{\text{seq}}, :\right] &= \mathbf{F}_{\text{gate}} \\ \mathbf{F}_{\text{restored}} &= \text{Reshape}\left(\mathbf{X}_{\text{flat}}\right) \in \mathbb{R}^{p \times p \times c} \end{aligned} \tag{11}$$

where the selected tokens are placed back at their original spatial positions, while the unselected pixels retain their original feature representations.

In parallel, to compensate for the limited ability of pure state-space modeling to capture high-frequency textures, we introduce a CNN-based local texture path. The local patch feature $\mathbf{X}_{\text{local}} = \mathbf{X}_{\text{p}} \in \mathbb{R}^{p \times p \times c}$ is processed by two lightweight convolutional layers with a SiLU activation

$$\mathbf{F}_{\text{cnn}} = \text{Conv}_2\left(\text{SiLU}\left(\text{Conv}_1\left(\mathbf{X}_{\text{local}}\right)\right)\right) \in \mathbb{R}^{p \times p \times c}. \tag{12}$$

The restored global semantic feature $\mathbf{F}_{\text{restored}}$ and local texture feature $\mathbf{F}_{\text{cnn}}$ now share the same spatial dimensions and are integrated through residual addition

$$\mathbf{F}_{\text{mid}} = \mathbf{F}_{\text{restored}} + \mathbf{F}_{\text{cnn}} \tag{13}$$

where $\mathbf{F}_{\text{mid}} \in \mathbb{R}^{p \times p \times c}$ denotes the intermediate fused feature combining global semantic and local texture information.

Finally, a residual postprocessing unit is employed to further refine the fused representation and stabilize feature distribution. Concretely, the fused feature is normalized and

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transformed by a pointwise convolution followed by a nonlinearity, and then added back to $\mathbf{F}_{\text{mid}}$ to yield the final global semantic feature

$$\mathbf{F}_{\text{global}} = \mathbf{F}_{\text{mid}} + \text{ReLU}\left(\text{PWConv}\left(\text{BN}\left(\mathbf{F}_{\text{mid}}\right)\right)\right) \tag{14}$$

where $\mathbf{F}_{\text{global}} \in \mathbb{R}^{p\times p\times c}$ represents the final output of the GSPB. This design enables LRMB to combine semantically consistent global modeling with boundary-aware local refinement in a compact and efficient manner.

3) Low-Rank SSM: HSI typically exhibits strong redundancy, and their spectral–spatial features are often well-characterized by a compact low-dimensional structure. Motivated by this property, we develop LRSSM that explicitly introduces a low-rank inductive bias into the core state evolution of Mamba. This design enables long-range dependency modeling with significantly fewer redundant parameters while enhancing robustness against irrelevant spectral variations.

Following the CASS, the neighborhood of each center pixel is organized as a semantically consistent sequence $\mathbf{X}_{\text{seq}} \in \mathbb{R}^{k\times c}$, where k denotes the sequence length and each token $\mathbf{x}_t \in \mathbb{R}^c$ corresponds to the feature of a selected homogeneous pixel. LRSSM maintains a hidden state $\mathbf{h}_t \in \mathbb{R}^N$ to accumulate contextual information along the scanning path, where N denotes the hidden state dimension. Accordingly, the standard SSM parameterization can be expressed as $\mathbf{A} \in \mathbb{R}^{N\times N}, \mathbf{B} \in \mathbb{R}^{N\times c}, \mathbf{C} \in \mathbb{R}^{c\times N}$.

To address the parameter redundancy in standard full-rank SSM, we constrain the evolution matrix $\mathbf{A}$ to a low-rank form by decomposing it into the product of a state coefficient matrix $\mathbf{U}_A \in \mathbb{R}^{N\times r}$ and a manifold basis matrix $\mathbf{V}_A \in \mathbb{R}^{N\times r}$

$$\mathbf{A} = \mathbf{U}_A \mathbf{V}_A^T \tag{15}$$

where $r \ll N$ is the preset manifold rank. $\mathbf{V}_A^T$ is responsible for projecting the hidden state from the previous time step onto the low-dimensional manifold subspace to extract evolutionary principal components, while $\mathbf{U}_A$ maps the evolved manifold features back to the system state space. This mechanism introduces a low-rank structural constraint on the state transition, encouraging the hidden dynamics to be dominated by a compact latent subspace.

Similarly, we also apply low-rank parameterization to the input injection and output readout matrices

$$\mathbf{B} = \mathbf{U}_B \mathbf{V}_B^\top, \quad \mathbf{C} = \mathbf{U}_C \mathbf{V}_C^\top \tag{16}$$

where $\mathbf{V}_B \in \mathbb{R}^{c\times r}$ acts as a compression operator that projects the c-dimensional input token into an r-dimensional latent representation to filter out input noise, before it is injected into the state space by $\mathbf{U}_B \in \mathbb{R}^{N\times r}$. Conversely, $\mathbf{V}_C \in \mathbb{R}^{N\times r}$ is responsible for extracting the most informative evolutionary components from the hidden state, while $\mathbf{U}_C \in \mathbb{R}^{c\times r}$ reconstructs these manifold features back to the original spectral space.

To incorporate Mamba-style selectivity, LRSSM employs an input-dependent timescale Δ_t generated from $\mathbf{x}_t$. Using zero-order hold (ZOH) discretization and a first-order approximation, the dynamic discrete operators are derived as $\bar{\mathbf{A}}_t = \exp(\Delta_t \mathbf{A})$ and $\bar{\mathbf{B}}_t \approx \Delta_t \mathbf{B} = \Delta_t \mathbf{U}_B \mathbf{V}_B^\top$. This explicitly preserves the input-dependent scaling on the injection path, consistent with selective state-space modeling. LRSSM performs recursion along the CASS-constructed sequence for $t = 1, \ldots, k$

$$\mathbf{h}_t = \bar{\mathbf{A}}_t \mathbf{h}_{t-1} + \bar{\mathbf{B}}_t \mathbf{x}_t \tag{17}$$

$$\mathbf{y}_t = \mathbf{C}\mathbf{h}_t + \mathbf{D}\mathbf{x}_t \tag{18}$$

where $\mathbf{D}$ is fixed as an identity matrix, corresponding to a residual feed-through connection.

CASS and LRSSM are complementary within the GSPB. CASS determines what to scan by selecting semantically related neighbors according to center-neighbor content affinity, thereby reducing heterogeneous token contamination before state-space modeling. LRSSM determines how to model the scanned sequence by factorizing the core SSM matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, which constrains hidden state dynamics within a compact latent subspace and reduces redundant full-rank responses. Thus, their joint use enables LRCS-Mamba to address spatial–semantic misalignment at the input-sequence level and spectral–spatial redundancy at the state-modeling level.

D. ADSG Module

To effectively integrate the complementary representations learned by the dual-branch architecture, we develop an ADSG module. As shown in Fig. 3, ADSG dynamically regulates the contributions of local texture cues from the local branch and global contextual semantics from the global branch in a content-adaptive manner, while promoting selective fusion by truncating low-contribution responses. Let $\mathbf{F}_{\text{local}}$ and $\mathbf{F}_{\text{global}}$ denote the feature maps produced by the local and global branches, respectively. These features are first concatenated and passed through a lightweight gating network to generate the intermediate gating feature $\mathbf{F}_{\text{att}}$

$$\mathbf{F}_{\text{att}} = \text{PW}_2\left(\text{ReLU}\left(\text{PW}_1\left(\text{BN}\left(\text{Concat}\left(\mathbf{F}_{\text{local}}, \mathbf{F}_{\text{global}}\right)\right)\right)\right)\right). \tag{19}$$

Subsequently, the branch-wise attention maps are generated by applying a Softmax operation to $\mathbf{F}_{\text{att}} \in \mathbb{R}^{p\times p\times 2}$

$$\left[\mathbf{G}_{\text{local}}, \mathbf{G}_{\text{global}}\right] = \text{Softmax}\left(\mathbf{F}_{\text{att}}\right) \tag{20}$$

where the Softmax is applied along the branch dimension, encouraging competitive weighting between the two feature streams. To obtain sparse features, ADSG employs threshold-based sparsification for both the local and global branches as follows:

$$\begin{aligned} \hat{\mathbf{G}}_{\text{local}} &= \mathbf{G}_{\text{local}} \odot \mathbb{I}\left(\mathbf{G}_{\text{local}} > \alpha\right) \\ \hat{\mathbf{G}}_{\text{global}} &= \mathbf{G}_{\text{global}} \odot \mathbb{I}\left(\mathbf{G}_{\text{global}} > \beta\right) \end{aligned} \tag{21}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function and $\odot$ denotes element-wise multiplication. Here, α and β are learnable threshold parameters for the local and global branches, respectively. To enable gradient-based optimization, the above hard thresholding is implemented differently during training and inference.

During the training stage, the hard thresholding is replaced by a continuous sigmoid relaxation to avoid the

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TABLE I
DETAILS OF EACH LAND-COVER CLASS OF FOUR DATASETS USED FOR EVALUATION IN THE EXPERIMENTS

No.	Indian Pines			University of Pavia			WHU-Hi-LongKou			Houston University		
	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing
1	Alfalfa	5	41	Asphalt	5	6626	Corn	5	34506	Healthy_grass	15	1359
2	Corn-notill	5	1423	Meadows	5	18644	Cotton	5	8369	Stressed_grass	15	1439
3	Corn-mintill	5	825	Gravel	5	2094	Sesame	5	3026	Synthetic_grass	15	745
4	Corn	5	232	Trees	5	3059	Broad-leaf soybean	5	63207	Trees	15	1249
5	Grass-pasture	5	478	Metal-sheets	5	1340	Narrow-leaf soybean	5	4146	Soil	15	1283
6	Grass-trees	5	725	Bare-soil	5	5024	Rice	5	11849	Water	15	310
7	Grass-pasture-m	5	23	Bitumen	5	1325	Water	5	67051	Residential	15	1461
8	Hay-windrowed	5	473	Bricks	5	3677	Roads and houses	5	7119	Commercial	15	1339
9	Oats	5	15	Shadows	5	942	Mixed weed	5	5224	Road	15	1539
10	Soybean-notill	5	967							Highway	15	1409
11	Soybean-mintill	5	2450							Railway	15	1498
12	Soybean-clean	5	588							Parking_lot_1	15	1414
13	Wheat	5	200							Parking_lot_2	15	555
14	Woods	5	1260							Tennis_court	15	466
15	Buildings-g-t-d	5	381							Running_track	15	743
16	Stone-steel-t	5	88									

nondifferentiability of the indicator function. Specifically, (21) is reformulated as

$$\hat{\mathbf{G}}_{\text{local}} = \mathbf{G}_{\text{local}} \odot \text{Sigmoid}\left(\frac{\mathbf{G}_{\text{local}} - \alpha}{\tau(t)}\right)$$
$$\hat{\mathbf{G}}_{\text{global}} = \mathbf{G}_{\text{global}} \odot \text{Sigmoid}\left(\frac{\mathbf{G}_{\text{global}} - \beta}{\tau(t)}\right) \tag{22}$$

where $\tau(t)$ is the temperature parameter controlling the sharpness of the continuous relaxation. We employ an exponential temperature annealing strategy

$$\tau(t) = \max\left(\tau_{\min}, \tau_0 \exp(-\gamma t)\right) \tag{23}$$

where $\tau_0 = 0.5$, $\tau_{\min} = 0.1$, $\gamma = 0.04$, and t denotes the epoch index starting from 0. The thresholds α and β are learnable parameters jointly optimized with the network through backpropagation. Both thresholds are obtained through a sigmoid transformation to constrain their values to $(0, 1)$ and are initialized to 0.5.

During the inference stage, the continuous relaxation is removed, and the general threshold-based formulation is implemented using the converged thresholds α^* and β^*

$$\hat{\mathbf{G}}_{\text{local}} = \mathbf{G}_{\text{local}} \odot \mathbb{I}(\mathbf{G}_{\text{local}} > \alpha^*)$$
$$\hat{\mathbf{G}}_{\text{global}} = \mathbf{G}_{\text{global}} \odot \mathbb{I}\left(\mathbf{G}_{\text{global}} > \beta^*\right). \tag{24}$$

Finally, the fused representation is obtained via gated feature aggregation

$$\mathbf{F}_{\text{fused}} = \left(\mathbf{F}_{\text{local}} \odot \hat{\mathbf{G}}_{\text{local}}\right) \oplus \left(\mathbf{F}_{\text{global}} \odot \hat{\mathbf{G}}_{\text{global}}\right) \tag{25}$$

where $\oplus$ denotes element-wise addition. Through content-adaptive branch competition and threshold-based sparsification, ADSG selectively preserves high-contribution cues from each branch, yielding a more discriminative fusion of local details and global semantics for HSI classification.

IV. EXPERIMENTAL RESULTS

A. Dataset Description

To comprehensively evaluate the performance of our proposed LRCS-Mamba method, we conducted experiments on four widely used and diverse HSI datasets: Indian Pines (IP), University of Pavia (UP), WHU-Hi-LongKou (LK), and Houston University 2013 (HU). These datasets represent a variety of remote sensing scenarios with different spatial resolutions, spectral characteristics, and land-cover types. Detailed information about the datasets is shown in Table I.

1) Indian Pines: The IP dataset was collected by the AVIRIS sensor in northwestern Indiana. It contains 220 spectral bands covering the wavelength range of 0.4–2.5 μm, with a spatial dimension of 145 $\times$ 145 pixels and 16 land-cover categories. After removing noisy bands, 200 spectral bands are retained.

2) University of Pavia: The UP dataset was captured by the reflective optics system imaging spectrometer (ROSIS) sensor over Pavia University in northern Italy. It consists of 610 $\times$ 340 pixels with a spatial resolution of 1.3 m/pixel and contains 103 spectral bands.

3) WHU-Hi-LongKou: The LK dataset was acquired using an autonomous aerial vehicle (AAV) equipped with a hyperspectral sensor over an agricultural zone in Hubei province, China. The image contains 550 $\times$ 400 pixels with 270 spectral bands ranging from 0.4 to 1.0 μm. The ground truth covers nine land cover classes, primarily consisting of agricultural areas and natural vegetation.

4) Houston University 2013: The HU dataset was collected by the Compact Airborne Spectrographic Imager (CASI-1500) over the University of Houston campus and its surrounding areas in Houston, Texas, with a spatial resolution of 2.5 m. The image has 349 $\times$ 1905 pixels and 144 spectral bands ranging from 0.38 to 1.05 μm. The ground truth includes 15 urban and suburban land cover categories.

B. Experimental Setups

1) Implementation Details: Both the comparative methods and LRCS-Mamba are trained using the Adam optimizer, with the learning rate set to 5×10^{-4} and the number of training epochs fixed at 200. The experiments were executed in Python 3.11 and PyTorch 2.2.2 with CUDA 12.1, and the experimental environment consists of an Intel Xeon Gold 6248R CPU and an NVIDIA Tesla V100S (32 GB) GPU. The partitioning of labeled samples into training and test sets is summarized

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TABLE II
QUANTITATIVE RESULTS (ACC% ± STD%) OF IP. BEST IN BOLD AND SECOND WITH UNDERLINE

Class	CNN-based		Transformer-based			Mamba-based				
	HPDM-SPRN	CSCANet	MLSFormer	MSFI-CNet	MViT	3DSS-Mamba	S^{2}Mamba	DBMGNet	MambaLG	LRCS-Mamba
1	99.76±0.73	91.66±8.33	**100.00±0.00**	99.68±1.57	**100.00±0.00**	**100.00±0.00**	97.56±1.15	**100.00±0.00**	95.71±1.42	**100.00±0.00**
2	64.43±7.44	45.47±1.38	72.52±1.97	53.92±8.55	59.65±5.23	69.11±3.91	70.56±8.23	69.43±5.87	**81.68±4.05**	76.04±3.14
3	62.99±9.97	81.31±0.42	61.37±12.61	**82.32±6.15**	63.75±3.87	59.51±4.32	67.76±3.66	62.28±4.21	70.81±8.54	50.06±5.07
4	85.04±8.30	98.01±0.66	97.27±2.99	94.42±2.12	**100.00±0.00**	96.92±0.82	97.41±2.15	98.25±0.55	94.24±3.98	97.84±1.06
5	89.73±4.08	**94.16±0.31**	81.94±2.86	92.07±1.67	74.53±2.95	87.32±1.91	82.85±2.56	92.01±1.38	83.89±1.05	82.01±0.62
6	**97.97±2.74**	96.79±0.69	90.53±2.61	93.11±2.43	93.35±1.29	97.36±0.82	85.52±5.59	97.78±0.69	93.46±1.39	90.34±1.42
7	99.13±1.74	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	97.05±2.94	**100.00±0.00**
8	92.47±7.85	97.20±2.36	98.52±1.13	99.25±0.57	99.15±0.49	99.63±0.23	97.67±1.05	84.68±6.15	84.36±13.27	**99.79±0.26**
9	**100.00±0.00**	95.00±4.99	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	71.31±8.03	77.69±2.88	73.15±16.79	71.69±4.63	77.28±3.58	73.65±4.42	81.59±8.87	64.41±7.08	74.76±5.77	**83.97±0.21**
11	58.65±8.32	76.68±2.25	79.81±4.86	67.20±9.75	72.21±6.54	79.67±2.77	75.43±5.17	77.07±3.01	62.82±4.72	**88.78±4.11**
12	75.20±12.35	78.06±2.07	75.91±11.26	59.59±16.24	78.80±2.53	64.15±4.09	**90.31±2.05**	79.14±2.81	64.69±8.16	80.61±3.99
13	99.50±0.32	98.97±1.04	99.83±0.24	**100.00±0.00**	99.49±0.34	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.96±0.57	99.97±0.03
14	88.90±7.06	93.80±1.16	98.32±0.62	83.90±3.36	91.33±1.37	92.67±2.32	96.43±5.40	85.68±2.08	89.19±2.75	**98.49±0.94**
15	80.29±10.03	95.56±1.74	65.88±7.08	89.79±2.29	**100.00±0.00**	89.10±1.62	69.29±8.18	86.24±4.01	90.13±3.46	78.74±6.79
16	97.50±4.57	98.19±0.62	**100.00±0.00**	98.88±1.11	92.94±1.58	96.39±1.97	98.86±0.54	**100.00±0.00**	99.39±0.61	**100.00±0.00**
OA	74.43±2.91	80.08±1.55	81.11±1.52	75.83±2.48	78.43±2.39	80.40±2.71	81.41±3.57	78.93±2.85	77.89±2.18	**84.67±1.52**
AA	85.18±2.19	88.66±1.44	87.19±1.73	86.68±2.55	87.66±1.75	87.71±1.86	88.20±2.68	87.31±1.24	86.32±4.03	**89.17±1.26**
BF1	68.52±3.24	76.14±1.78	78.34±2.89	71.88±4.38	73.56±2.12	77.92±2.35	79.45±3.41	74.08±2.56	73.12±2.92	**83.64±1.38**
Kappa	71.19±3.21	77.50±2.62	78.75±3.55	72.91±3.17	75.75±2.41	77.68±2.97	78.91±3.48	76.12±3.18	73.78±2.31	**82.57±1.92**

in Table I. For each dataset, samples are randomly divided into mutually exclusive training and test sets according to the specified counts. No separate validation set is used. A different random seed is used for each experimental run, resulting in a distinct data split for every trial.

2) Evaluation Metrics: Five commonly used evaluation metrics, including overall accuracy (OA), average accuracy (AA), category accuracy (CA), boundary $F1$-score (BF1), and the Kappa coefficient, were employed to evaluate the classification performance of the proposed method. Among them, BF1 is introduced to measure the consistency between predicted boundaries and ground-truth boundaries, providing a more direct evaluation of boundary preservation capability. A higher BF1 value indicates better consistency between predicted and ground-truth boundaries. To eliminate the impact of randomness, all experiments were performed ten times, and the mean and standard deviation values of each evaluation metric were reported.

3) Baseline: To evaluate the performance of LRCS-Mamba, we compared it with nine advanced HSI classification methods, including two CNN-based methods (HPDM-SPRN [52] and CSCANet [53]), three Transformer-based methods (MLSFormer [50], MSFI-CNet [54], and MViT [55]), and four Mamba-based methods (3DSS-Mamba [6], S^{2}Mamba [36], DBMGNet [42], and MambaLG [2]). Among these methods, MLSFormer and MSFI-CNet explicitly incorporate low-rank and sparse modeling strategies into Transformer-based feature representation, providing additional comparisons with methods that exploit low-rank and sparse priors.

C. Classification Results

1) Quantitative Analysis: Tables II–V present a comprehensive quantitative comparison between LRCS-Mamba and nine state-of-the-art baselines on the IP, UP, LK, and HU datasets. The compared methods cover CNN-based, Transformer-based, low-rank and sparse-prior Transformer-based, and Mamba-based models, providing a more complete evaluation of different spectral–spatial modeling paradigms. In addition to OA, AA, and Kappa, we further report BF1 to evaluate boundary preservation capability. Overall, LRCS-Mamba consistently achieves the best OA, AA, BF1, and Kappa on all four datasets, demonstrating strong classification performance across different HSI scenes.

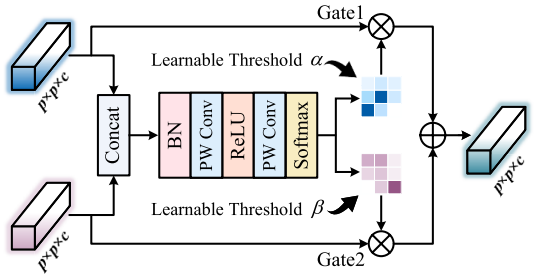


Fig. 3. ADSG module.

Building upon this advantage, LRCS-Mamba further improves spectral–spatial representation by integrating CASS and low-rank state-space modeling. On the IP dataset, LRCS-Mamba achieves 84.67% OA, 89.17% AA, and 82.57% Kappa, outperforming the corresponding second-best results by 3.26%, 0.51%, and 3.66%, respectively, which indicates its effectiveness in heterogeneous agricultural scenes with limited training samples. Similar advantages can be observed on UP, where LRCS-Mamba obtains 88.93% OA, 88.96% AA, and 85.46% Kappa, showing improved robustness under complex urban class-mixing conditions. For LK, LRCS-Mamba reaches 95.73% OA, 93.78% AA, and 94.44% Kappa, demonstrating strong generalization in large-scale agricultural scenes. On HU, it achieves 94.14% OA, 94.64% AA, and 92.84% Kappa, further confirming its effectiveness for fine-grained classification in complex urban environments.

Although LRCS-Mamba achieves the best OA on all four datasets, it does not obtain the highest accuracy for every individual class. For example, on the IP dataset, the accuracies

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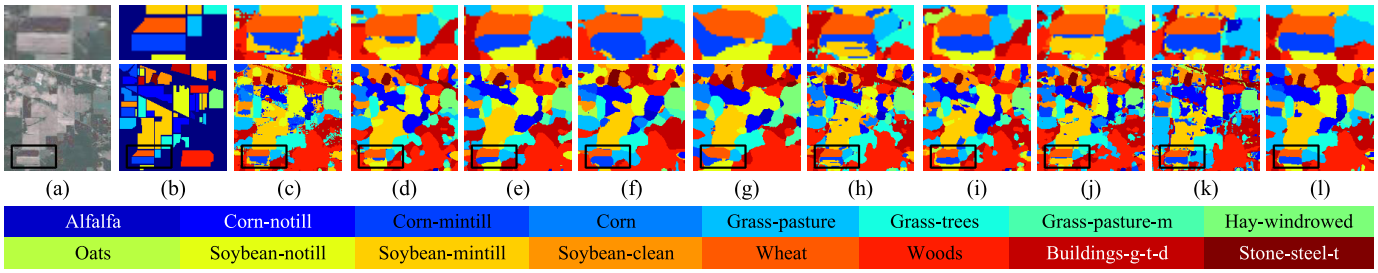


Fig. 4. Classification maps of different methods on the IP dataset. (a) False-color image. (b) Ground truth. (c) HPDM-SPRN. (d) CSCANet. (e) MLSFormer. (f) MSFI-CNet. (g) MViT. (h) 3DSS-Mamba. (i) S^{2}Mamba. (j) DBMGNet. (k) MambaLG. (l) LRCS-Mamba.

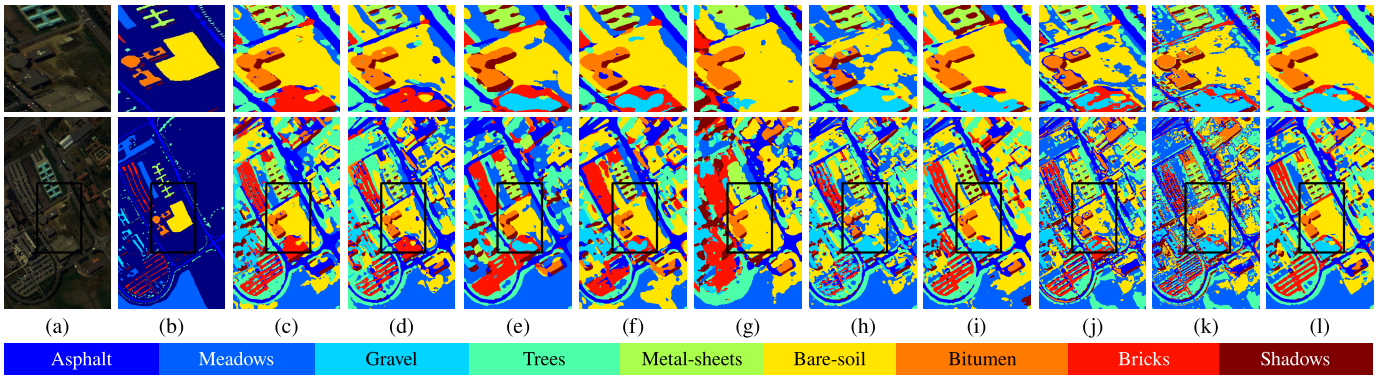


Fig. 5. Classification maps of different methods on the UP dataset. (a) False-color image. (b) Ground truth. (c) HPDM-SPRN. (d) CSCANet. (e) MLSFormer. (f) MSFI-CNet. (g) MViT. (h) 3DSS-Mamba. (i) S^{2}Mamba. (j) DBMGNet. (k) MambaLG. (l) LRCS-Mamba.

of classes 3 and 5 are lower than those of several competing methods. This may be attributed to the stronger intraclass spectral variability of these categories. Under the fixed sparse ratio used in CASS, the selected homogeneous neighboring pixels may not fully represent the diversity of such heterogeneous classes, and some useful but spectrally diverse contextual pixels may be excluded. Therefore, the fixed sparse scanning strategy provides a favorable global tradeoff, but may be suboptimal for certain highly heterogeneous land-cover classes.

To comprehensively evaluate boundary robustness, we further report BF1 in Tables II–V. Unlike OA and AA, which mainly reflect region-level classification accuracy, BF1 focuses on the consistency between predicted and ground-truth boundaries and is more sensitive to boundary discontinuities and mixed-pixel misclassification. LRCS-Mamba achieves the highest BF1 values on all four datasets, with 83.64%, 86.38%, 93.93%, and 94.01% on IP, UP, LK, and HU, respectively. Compared with the corresponding second-best methods, the BF1 improvements are 4.19%, 4.31%, 2.37%, and 1.56%, respectively. These results provide direct quantitative evidence that LRCS-Mamba can better preserve class boundaries and improve boundary-sensitive classification performance.

Overall, the comparison with low-rank and sparse-prior baselines and the added BF1 results further verify the effectiveness of LRCS-Mamba. Meanwhile, the class-wise observations suggest that adaptive or class-aware sparse scanning may further improve performance for highly heterogeneous land-cover categories.

2) Visual Evaluation: Figs. 4–7 visualize the classification maps of LRCS-Mamba and representative baselines on IP, UP, LK, and HU, where the black boxes denote enlarged regions for detailed inspection. Overall, LRCS-Mamba yields more spatially coherent regions with fewer isolated misclassified pixels, while producing sharper class boundaries, indicating improved class integrity and reduced salt-and-pepper noise. On IP, the Corn-mintill area in the zoomed region forms more contiguous patches with less speckle noise. On UP, the boundary between Bitumen and surrounding classes shows noticeably less boundary bleeding, alleviating class-mixing artifacts. On LK, LRCS-Mamba better preserves the shape continuity of broad-leaf soybean, reflecting more accurate regional modeling in large-scale scenes. On HU, small structured targets remain more consistent despite spectral similarity to adjacent materials. These observations are consistent with the roles of CASS in mitigating boundary misalignment and low-rank state-space modeling in suppressing spectral noise, and they agree well with the quantitative results.

D. Hyperparameter Analysis

1) Influence of Patch Size: Fig. 8(a) reports the classification accuracy of LRCS-Mamba on four datasets with different patch sizes. A small patch size generally results in suboptimal performance, likely due to insufficient spatial context for modeling local spatial structures. Notably, the accuracy does not improve monotonically with larger patches, suggesting that overly large neighborhoods may introduce redundant background or mixed pixels that weaken class discrimination. In general, the optimal patch size is often related to the spatial resolution and the spatial homogeneity of land-cover distributions in HSI. For IP, LK, and HU, the best

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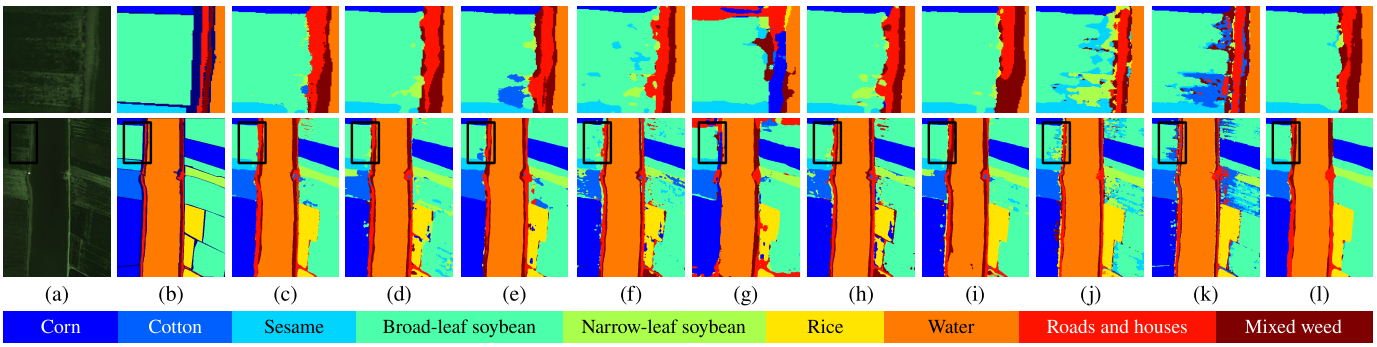


Fig. 6. Classification maps of different methods on the LK dataset. (a) False-color image. (b) Ground truth. (c) HPDM-SPRN. (d) CSCANet. (e) MLSFormer. (f) MSFI-CNet. (g) MViT. (h) 3DSS-Mamba. (i) S^{2}Mamba. (j) DBMGNet. (k) MambaLG. (l) LRCS-Mamba.

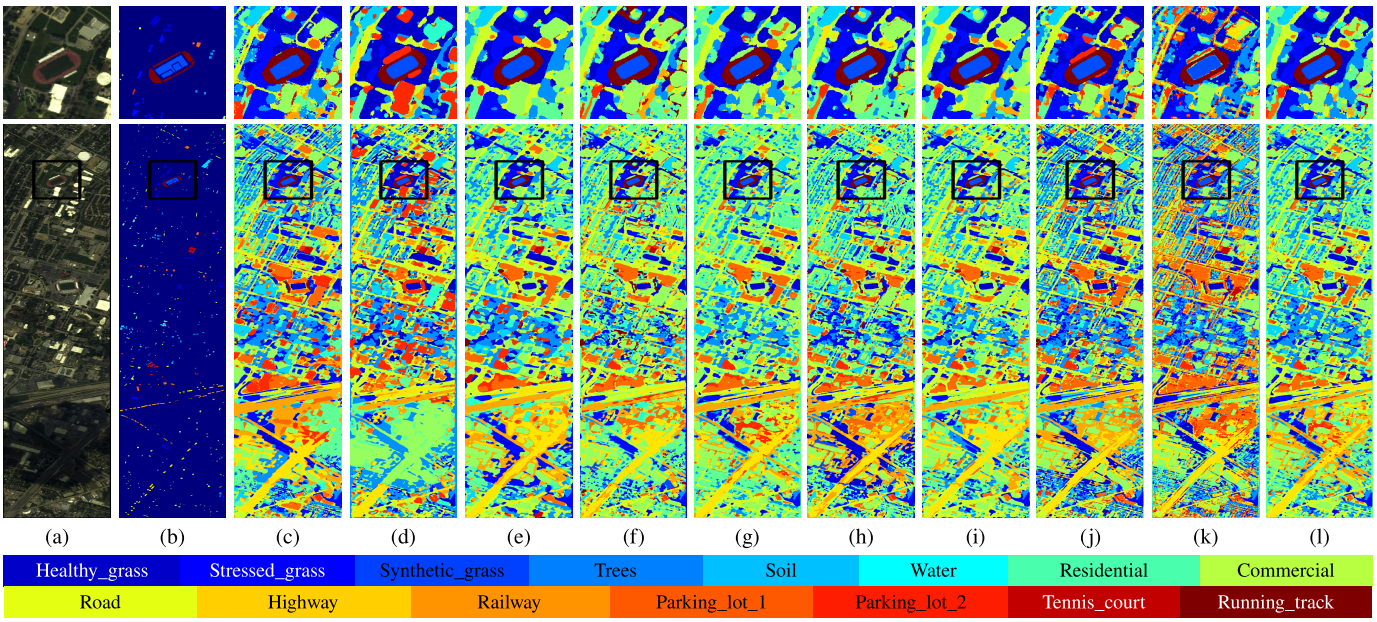


Fig. 7. Classification maps of different methods on the HU dataset. (a) False-color image. (b) Ground truth. (c) HPDM-SPRN. (d) CSCANet. (e) MLSFormer. (f) MSFI-CNet. (g) MViT. (h) 3DSS-Mamba. (i) S^{2}Mamba. (j) DBMGNet. (k) MambaLG. (l) LRCS-Mamba.

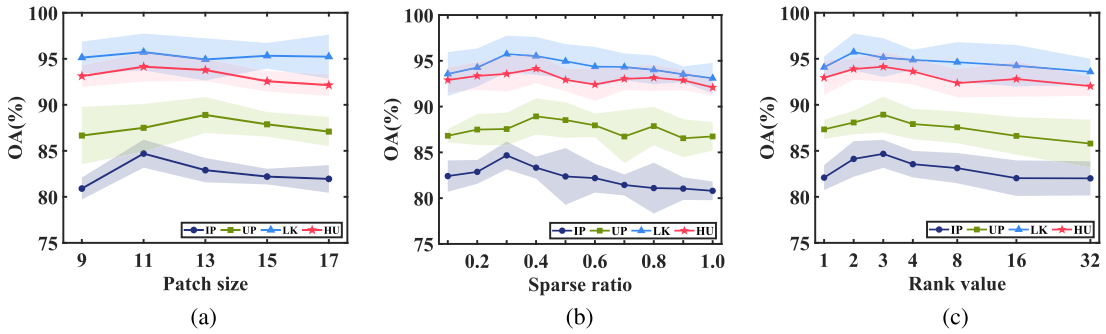


Fig. 8. Classification accuracies of the proposed LRCS-Mamba under different (a) patch sizes, (b) sparse ratios, and (c) rank values, where the shaded region denotes the standard deviation.

performance is achieved at a patch size of 11 $\times$ 11, indicating that this neighborhood is sufficient to capture essential spatial context. For UP, the optimum shifts to 13 $\times$ 13, which may be attributed to its higher spatial heterogeneity and denser boundaries. Nevertheless, to maintain a consistent and fair experimental protocol across datasets and to improve computational efficiency, we adopt a unified patch size of 11 $\times$ 11 for all experiments. This setting still yields consistently strong performance, indicating that LRCS-Mamba is not overly sensitive to the patch-size choice.

2) Influence of Sparse Ratio (s): Fig. 8(b) illustrates the classification performance of LRCS-Mamba under varying sparse ratios $s \in [0.1, 1.0]$, where the curves denote the mean

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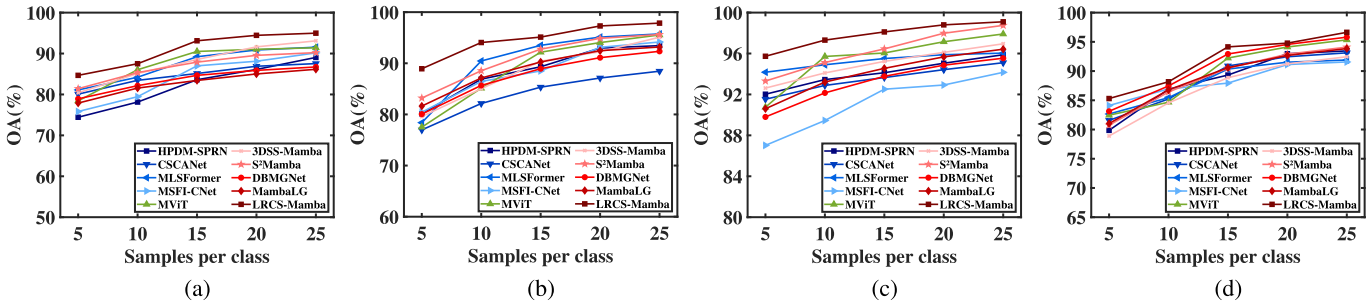


Fig. 9. Classification accuracies of different methods under different numbers of training samples on each dataset. (a) IP. (b) UP. (c) LK. (d) HU.

OA over repeated runs, and the shaded colored regions indicate the corresponding variance. The OA across all four datasets exhibits a consistent parabolic trend, with the accuracy initially rising toward an optimal value and subsequently declining as s approaches 1.0, where no sparsification is applied. This observation validates the core motivation of our CASS mechanism. Specifically, an overly low sparse ratio (e.g., 0.1) leads to significant information loss as critical homogeneous neighbors are excluded from the sequence. Conversely, as s increases beyond the optimal range of 0.3–0.4, the inclusion of heterogeneous or boundary pixels introduces semantic aliasing, which inherently degrades the semantically consistent path of Mamba's state evolution. When s approaches 1.0, the sequence becomes increasingly contaminated by irrelevant background noise, resulting in a noticeable performance drop. Notably, the IP and LK datasets reach their peak at $s = 0.3$, and the UP and HU datasets prefer $s = 0.4$. For experimental consistency and to maintain a favorable tradeoff across all scenes, s is fixed to 0.4 for all subsequent evaluations.

3) Influence of Rank Value (r): Fig. 8(c) summarizes the classification performance of LRCS-Mamba under different rank settings of the LRSSM operator, where the curves denote the mean OA over repeated runs, and the shaded colored regions indicate the corresponding variance. The results indicate that the rank configuration has a clear impact on model efficacy. As the rank increases, the accuracy typically improves initially, then saturates, and may slightly decline at overly large ranks. When the rank is too small (e.g., $r = 1$), the model suffers from a representation bottleneck, where the constrained state-space dynamics are insufficient to capture the complex spectral–spatial correlations inherent in HSI. In contrast, overly large ranks (e.g., $r \geq 8$) weaken the low-rank inductive bias, making it harder to distill coherent structural information and to suppress irrelevant spectral variations. Notably, the IP, UP, and HU datasets achieve their highest OAs at $r = 3$, whereas the LK dataset peaks at $r = 2$ due to its relatively simpler land-cover structure. Accordingly, the rank value is fixed to 3 across all datasets to ensure a consistent and efficient modeling configuration.

E. Influence of Training Set Size

We evaluate the classification performance of the proposed method with different numbers of training samples. Experiments are conducted on four datasets, where the number of training samples per class is set to [5, 10, 15, 20, 25]. Fig. 9 shows the OA of different methods as the number of training samples per class varies.

In general, the OA of all methods increases as more labeled samples are available. It is worth noting that LRCS-Mamba remains robust under limited training samples. As shown in Fig. 9, with only five training samples per class, LRCS-Mamba achieves approximately 84.67%, 88.93%, 95.73%, and 85.31% OA on IP, UP, LK, and HU, respectively. This advantage mainly comes from the joint effects of CASS and LRSSM. CASS shifts noise handling from data-driven suppression to structure-driven filtering through content-aware Top-k scanning, thereby reducing heterogeneous token contamination before state-space modeling. Meanwhile, LRSSM constrains the Mamba branch within a compact low-rank subspace, reducing overfitting to small-sample spectral fluctuations. Moreover, the prefiltering effect of CASS makes this low-rank constraint more reliable, encouraging LRSSM to model discriminative structures rather than residual noise. Thus, CASS improves input reliability, while LRSSM improves generalization under few-shot HSI classification. When the number of training samples per class increases to 10, 15, 20, and 25, LRCS-Mamba consistently maintains the best classification performance. Similar trends are observed on the UP, LK, and HU datasets. These results indicate that the proposed method remains effective under different training-set sizes, especially when labeled samples are scarce, which is particularly important for HSI classification where acquiring labeled data is expensive.

F. Ablation Study

To evaluate the contribution of each component in LRCS-Mamba, we conduct ablation experiments on four datasets, and the results are summarized in Table VI. Cases 1and 2 evaluate two single-branch variants: Case 1 keeps only the LTPB, while Case 2 keeps only the Mamba-based semantic branch composed of CASS and LRMB. Cases 3–5 perform component-wise ablations on the dual-branch architecture: Case 3 removes ADSG, and Case 4 replaces CASS with DefMamba [7], a recent nongrid deformable scanning strategy that adaptively adjusts scanning paths according to geometric offsets. Case 5 replaces LRMB with a standard Mamba block, and Case 6 denotes the complete LRCS-Mamba.

As shown in Table VI, both single-branch variants perform worse than the complete model on all datasets. Compared with Case 6, Case 1 yields OA drops of 3.32%, 4.81%, 3.58%, and

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TABLE III

QUANTITATIVE RESULTS (ACC% ± STD%) OF UP. BEST IN BOLD AND SECOND WITH UNDERLINE

Class	CNN-based		Transformer-based			Mamba-based				
	HPDM-SPRN	CSCANet	MLSFormer	MSFI-CNet	MViT	3DSS-Mamba	S^2Mamba	DBMGNet	MambaLG	LRCS-Mamba
1	84.83±4.30	61.08±2.15	81.31±12.93	64.34±6.64	56.76±4.13	72.55±4.19	81.95±2.46	74.10±3.37	**91.84±0.31**	63.69±6.12
2	66.69±10.59	87.57±3.39	65.58±15.58	74.34±4.11	80.28±2.67	77.87±3.16	75.44±3.25	73.98±3.39	76.04±0.79	**93.96±2.51**
3	86.55±10.13	69.76±2.15	78.63±9.95	66.83±2.65	73.16±3.58	62.72±9.18	69.72±3.85	79.77±2.76	**95.30±1.53**	68.62±4.16
4	94.67±2.55	80.74±3.73	90.17±3.11	94.15±3.32	78.59±2.84	95.13±3.77	92.70±1.32	**95.84±0.95**	93.12±1.17	92.19±3.48
5	99.95±0.08	94.18±3.64	98.73±1.27	99.78±0.22	**100.00±0.00**	99.78±0.55	91.85±1.39	**100.00±0.00**	99.94±0.03	99.96±0.06
6	90.59±13.94	62.95±6.70	95.67±4.48	99.98±0.02	98.77±0.46	81.24±3.46	**100.00±0.00**	82.91±3.38	76.06±5.34	88.85±3.96
7	99.69±0.51	85.54±8.39	97.76±0.92	99.02±0.27	**100.00±0.00**	95.69±1.64	94.98±1.09	90.46±1.48	95.75±1.36	99.77±0.23
8	90.48±3.95	61.23±1.54	86.50±6.70	94.54±2.18	58.50±3.92	77.25±5.44	94.64±1.13	88.94±2.60	65.94±2.94	**99.43±0.11**
9	**99.50±0.55**	83.86±6.73	92.47±4.46	86.64±2.27	72.19±3.61	99.47±0.09	81.30±2.57	98.72±0.49	98.07±0.42	88.22±5.41
OA	80.13±5.94	76.97±1.42	78.39±5.42	80.43±2.51	77.51±3.09	79.59±1.03	83.20±2.34	80.06±2.73	81.63±1.23	**88.93±1.95**
AA	86.33±3.00	76.21±2.69	87.43±1.44	86.62±2.56	79.80±2.73	84.63±1.96	86.95±1.94	87.19±1.92	87.66±1.17	**88.96±1.16**
BF1	79.42±4.15	73.18±1.82	74.96±3.68	77.51±2.95	74.85±2.44	78.91±1.52	82.07±2.11	79.54±1.96	79.78±1.12	**86.38±1.42**
Kappa	75.31±7.01	69.91±1.72	73.33±6.04	75.49±3.49	71.57±3.79	73.91±1.35	82.47±2.41	74.76±3.31	76.56±1.72	**85.46±2.59**

TABLE IV

QUANTITATIVE RESULTS (ACC% ± STD%) OF LK. BEST IN BOLD AND SECOND WITH UNDERLINE

Class	CNN-based		Transformer-based			Mamba-based				
	HPDM-SPRN	CSCANet	MLSFormer	MSFI-CNet	MViT	3DSS-Mamba	S^2Mamba	DBMGNet	MambaLG	LRCS-Mamba
1	98.88±1.02	97.77±2.44	98.09±1.27	93.21±5.60	**99.28±0.31**	95.99±1.47	97.30±1.73	97.15±0.75	98.32±0.26	99.13±2.71
2	84.03±6.17	89.91±7.28	94.31±5.24	66.27±9.56	93.38±1.27	71.38±12.17	87.31±1.91	97.64±0.68	**97.96±1.59**	90.73±4.78
3	96.70±2.58	98.11±2.01	98.38±1.29	81.90±13.33	99.93±0.07	98.71±1.73	91.60±3.44	**99.96±0.06**	99.55±0.18	90.58±3.57
4	86.29±5.78	86.54±5.09	94.87±2.21	75.20±9.38	87.42±5.89	87.80±2.61	94.83±1.14	70.37±3.77	75.48±1.05	**96.81±3.11**
5	**99.86±0.14**	99.66±0.34	82.86±19.32	68.40±10.79	97.56±0.68	73.19±8.54	96.99±0.89	98.98±0.42	56.82±5.59	98.31±1.02
6	85.57±6.07	97.38±2.24	96.82±2.05	84.85±6.52	89.17±2.52	95.05±9.42	92.20±1.36	98.38±0.53	**98.99±0.49**	89.23±3.51
7	95.55±2.09	93.27±2.90	99.37±0.29	99.66±0.42	95.94±0.93	99.91±0.09	91.17±1.45	**99.99±0.01**	99.63±0.18	95.42±1.87
8	95.56±2.44	80.64±2.41	60.33±11.12	91.48±1.94	35.96±15.75	86.24±2.94	89.97±1.53	95.54±1.28	95.21±1.63	**95.72±2.30**
9	83.11±5.66	82.21±4.05	71.64±8.69	76.75±12.70	71.54±6.64	84.05±1.65	90.68±2.46	**92.20±1.36**	91.20±2.49	88.06±5.72
OA	92.02±3.87	91.53±3.69	94.17±1.19	87.02±2.93	90.75±3.14	92.62±0.94	93.31±1.26	89.80±2.55	90.60±1.58	**95.73±2.01**
AA	91.84±2.33	91.72±1.84	88.52±3.78	81.97±1.97	85.58±2.23	88.04±1.32	92.45±1.33	92.47±1.14	90.36±1.49	**93.78±2.11**
BF1	89.45±2.34	88.92±2.15	91.56±1.28	74.82±2.73	87.34±2.67	89.67±1.02	91.24±1.21	88.05±1.65	89.12±1.34	**93.93±1.85**
Kappa	89.74±2.41	89.10±4.58	92.45±0.65	83.45±3.57	88.04±2.67	90.43±1.20	93.18±1.49	87.04±1.93	87.98±3.60	**94.44±2.86**

TABLE V

QUANTITATIVE RESULTS (ACC% ± STD%) OF HU. BEST IN BOLD AND SECOND WITH UNDERLINE

Class	CNN-based		Transformer-based			Mamba-based				
	HPDM-SPRN	CSCANet	MLSFormer	MSFI-CNet	MViT	3DSS-Mamba	S^2Mamba	DBMGNet	MambaLG	LRCS-Mamba
1	95.85±5.97	83.64±4.53	93.72±4.31	95.87±1.33	98.30±0.54	**99.11±0.81**	93.23±5.69	93.28±1.26	98.13±0.57	96.12±1.78
2	94.90±4.56	94.78±0.20	93.57±2.41	93.70±2.54	94.27±1.18	87.41±4.10	**98.47±0.41**	96.61±0.89	86.02±2.02	91.53±3.14
3	**100.00±0.00**	98.08±0.87	98.58±1.28	96.77±1.36	99.85±0.15	**100.00±0.00**	99.80±0.07	**100.00±0.00**	98.53±0.53	**100.00±0.00**
4	97.15±2.06	89.24±2.24	92.76±0.18	92.51±4.11	86.42±1.97	93.08±2.25	97.21±0.93	**99.83±0.17**	95.52±1.27	96.01±1.36
5	**100.00±0.00**	99.91±0.08	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.92±0.39	99.48±0.44	**100.00±0.00**	98.04±0.68	99.67±0.56
6	96.69±5.54	98.05±1.29	93.33±3.78	88.06±5.28	88.42±2.63	87.10±5.94	95.70±6.08	98.06±0.59	**99.83±0.15**	87.10±5.43
7	92.42±2.27	90.66±3.31	89.49±1.64	87.23±3.45	90.19±1.51	86.99±3.31	90.02±4.33	91.77±1.42	82.54±2.49	**97.05±1.48**
8	66.31±10.58	64.01±1.57	73.09±1.02	74.04±6.17	83.82±2.24	75.43±5.95	66.67±1.11	78.51±2.86	83.75±3.54	**88.84±3.35**
9	**86.72±4.27**	86.48±2.97	86.11±2.64	83.02±2.56	78.43±5.88	83.19±4.73	85.69±3.03	81.89±2.48	82.66±2.39	86.26±4.42
10	84.22±11.01	95.25±1.73	86.83±9.43	78.47±2.34	**99.51±0.26**	85.97±5.44	89.00±5.14	93.39±1.26	89.21±1.54	97.52±2.37
11	87.05±2.31	97.82±1.61	90.77±3.65	87.70±3.21	96.81±0.87	92.62±5.41	92.46±4.16	**99.50±0.34**	84.18±2.17	89.67±1.70
12	69.36±13.80	**94.28±1.28**	83.85±8.69	71.84±5.64	85.97±2.19	67.16±7.51	76.79±9.27	82.84±2.38	89.56±1.56	90.07±2.32
13	94.27±3.95	90.04±0.66	88.84±4.01	86.34±3.48	93.19±1.27	95.37±1.16	93.61±1.25	93.61±1.23	96.57±0.86	**100.00±0.00**
14	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.76±0.20	**100.00±0.00**	**100.00±0.00**	99.63±0.23	**100.00±0.00**
15	**100.00±0.00**	99.22±0.31	99.22±0.79	**100.00±0.00**	97.06±0.78	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.21±0.56	99.69±0.54
OA	89.32±1.98	90.88±2.57	90.35±1.72	87.94±1.22	92.23±1.35	88.87±1.44	90.50±1.09	92.92±1.30	90.58±1.47	**94.14±1.60**
AA	91.00±1.75	92.10±1.41	91.38±1.66	89.04±1.69	92.82±1.31	90.21±1.12	91.88±1.11	93.96±1.19	92.16±1.37	**94.64±1.26**
BF1	88.54±1.65	89.76±2.12	88.49±2.89	88.89±1.74	91.15±1.24	87.42±1.35	89.28±1.02	92.45±1.22	89.92±1.38	**94.01±1.48**
Kappa	88.46±2.13	90.14±1.62	89.57±1.86	86.96±2.57	91.60±1.44	87.97±1.56	89.73±1.17	92.35±1.34	90.10±1.58	**92.84±0.65**

4.49% on IP, UP, LK, and HU, respectively, while Case 2 leads to corresponding drops of 2.11%, 3.85%, 4.59%, and 2.42%. These results indicate that the local texture branch and the semantic Mamba branch provide complementary information, and that combining them is beneficial for HSI classification. For component-wise ablations, replacing ADSG in Case 3 with a simple addition operation of the two branches decreases OA by 1.25%, 0.69%, 0.78%, and 0.78% on the four datasets, respectively. This suggests that adaptive gated fusion helps balance local details and semantic context more effectively than direct fusion. Replacing CASS with DefMamba scanning strategy in Case 4 causes OA reductions of 0.89%, 0.72%, 0.55%, and 0.50%, showing that CASS consistently contributes to constructing more effective token sequences.

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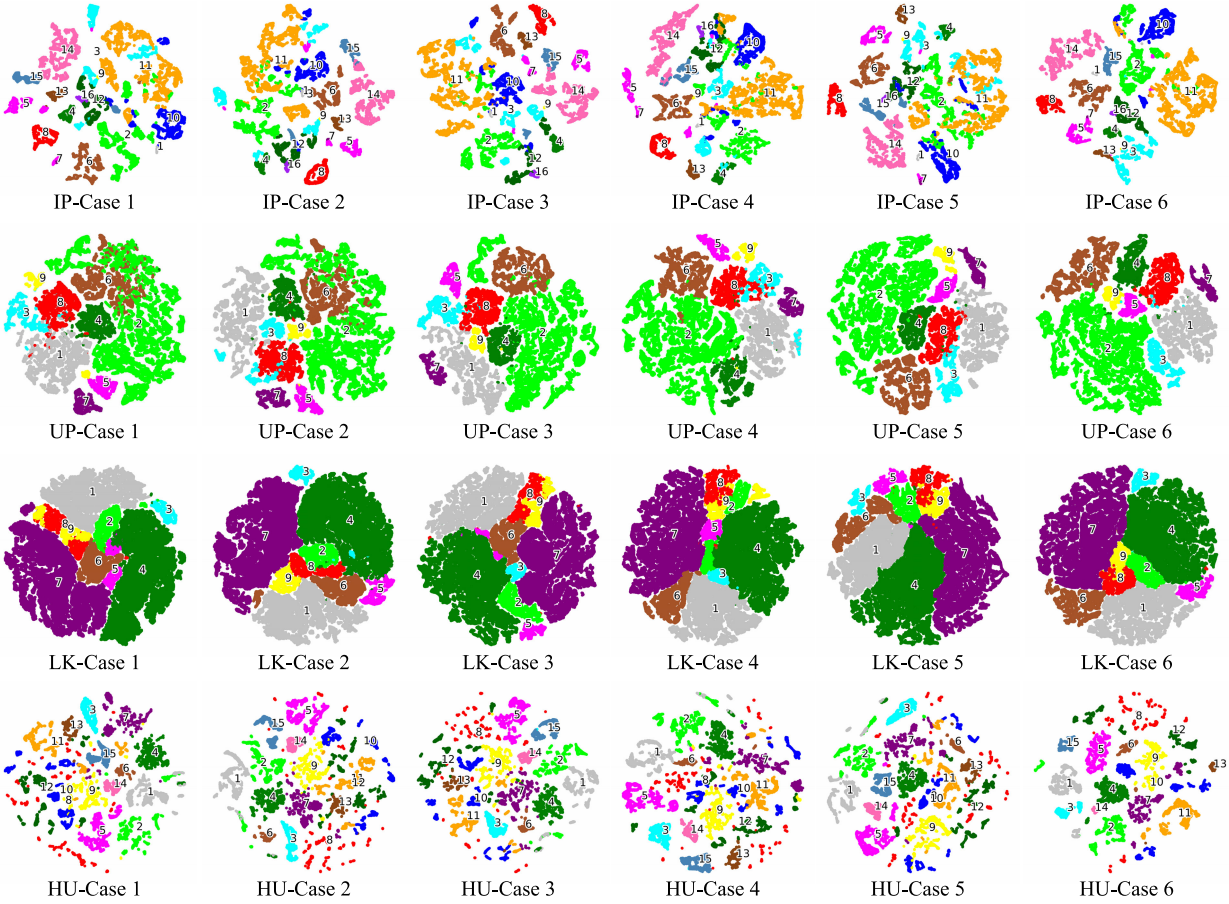


Fig. 10. t-SNE visualization of the ablation study. Each row from top to bottom corresponds to the IP, UP, LK, and HU datasets, respectively.

TABLE VI
ABLATION ANALYSIS ON FOUR DATASETS. BEST IN BOLD

Case	LTPB	GSPB		ADSG	Datasets			
		CASS	LRMB		IP	UP	LK	HU
1	✓	✗	✗	✗	81.35±1.72	84.12±2.19	92.15±3.27	89.65±1.98
2	✗	✓	✓	✗	82.56±1.49	85.08±1.78	91.14±2.11	91.72±1.82
3	✓	✓	✓	✗	83.42±1.80	88.24±1.27	94.95±2.23	93.36±1.62
4	✓	✗	✓	✓	83.78±1.42	88.21±1.65	95.18±1.76	93.64±1.81
5	✓	✓	✗	✓	84.12±1.61	88.35±1.48	95.42±1.94	93.71±1.38
6	✓	✓	✓	✓	**84.67±1.52**	**88.93±1.95**	**95.73±2.01**	**94.14±1.60**

Replacing LRMB with a standard Mamba block in Case 5 also leads to consistent performance degradation, with OA drops of 0.55%, 0.58%, 0.31%, and 0.43%, indicating that the low-rank design provides additional gains by encouraging compact spectral–spatial representation learning. Together, Cases 4 and 5 suggest that CASS improves semantic sequence construction, while LRMB enhances compact state-space modeling. Overall, these ablation results demonstrate that each proposed component contributes to the final performance, while their combination achieves the best results across all datasets.

To provide a qualitative view of feature separability, we further visualize the learned embeddings using t-SNE in Fig. 10. The complete model generally produces more compact intraclass distributions and clearer interclass separation than the ablated variants, which is consistent with the quantitative results in Table VI. In comparison, the embeddings of the ablated models show more overlap among visually or spectrally similar categories. These observations further suggest that the proposed CASS, LRMB, and ADSG components jointly improve the discriminability of learned spectral–spatial features.

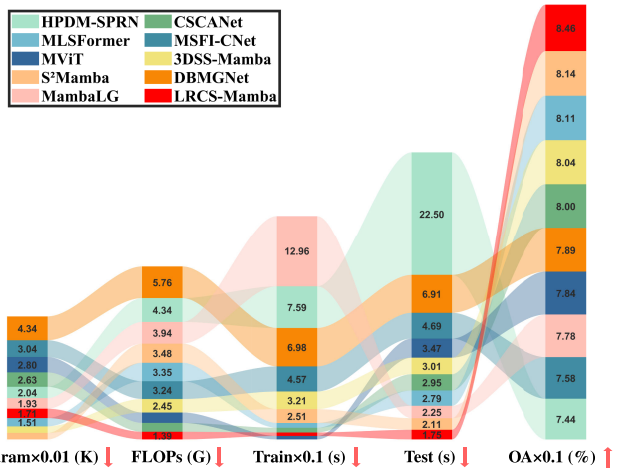


Fig. 11. Model complexity analysis of all methods on IP dataset.

G. Complexity Analysis

Computational complexity is a key factor for the practical applicability of HSI classifiers. We evaluate the comparison

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methods on the IP dataset in terms of parameter count, FLOPs, training time, and inference time. For a fair comparison, these metrics are measured by considering all processing stages of each method. For LRCS-Mamba, these metrics are measured over the complete pipeline, including the CASS-based sequence construction. As shown in Fig. 11, LRCS-Mamba achieves the lowest FLOPs (1.39G) and the fastest inference (1.75 s), indicating strong computational efficiency. In addition, LRCS-Mamba has a moderate number of parameters (171.83 K) and requires 6.46 s for training, while remaining substantially more efficient than Transformer-based baselines such as MSFI-CNet (304.59 K, 45.72 s) and competitive with MLSFormer (151.46 K, 8.73 s). Overall, LRCS-Mamba attains the highest OA with competitive computational cost, suggesting a good tradeoff between accuracy and efficiency.

V. Conclusion

In this article, we introduced LRCS-Mamba, a Mamba model that integrates CASS with an LRSSM for HSI classification. CASS alleviates spatial–semantic misalignment caused by rigid geometric scanning by selecting semantically related neighbors, while LRSSM factorizes core SSM operators to encourage state dynamics within a compact low-dimensional latent manifold. The proposed design relies on the assumptions that spectrally similar neighboring pixels are more likely to be semantically consistent and that HSI features can be effectively represented in a low-dimensional subspace. Experimental results demonstrate that incorporating domain-specific priors, such as semantic sparsity and low-rank manifolds, into Mamba improves model robustness and discriminative performance. From a practical perspective, LRCS-Mamba may be useful for HSI scenes with limited labeled samples, strong spectral redundancy, and irregular land-cover boundaries. Nevertheless, the sparse ratio in CASS and the rank value in LRSSM are currently fixed, while different scenes and land-cover categories may have different spatial distributions, spectral variability, and manifold complexity. Future work will therefore explore adaptive sparse scanning and rank estimation strategies, as well as more flexible topological constraints for large-scale remote sensing scenes.

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